



Fig. 2: With twelve modules and boards in the form factors COM Express, Mini-ITX, Flex-ATX and CPCI, AMC as well as VPX, there is already a wide standard assortment of latest solutions based on Intel or AMD processors. These solutions already have USB 3.0 integrated in the chipset and can be adapted to suit customers' specifications

more promising solution for a large range of new designs.

USB 3.0 now available in the chipset

The most significant cost factor in PC implementation of USB 3.0 is the availability in the chipset. Since the launch of third-generation Intel Core processors and AMD Embedded R-Series processors, the new interface standard USB 3.0 has been integrated into these manufacturers' chipsets and thus is available as a standard on embedded board and system level, which proves immense cost-saving.

USB 3.0 implementations by OEMs

Widespread introduction into the industrial environment has not yet taken place and—in terms of embedded PC technology—can be seen as still in its infancy, as the USB 3.0 chipsets have only just been launched and still have to be designed-in.

On the other hand, a relatively inexpensive way to realise this is not really available: At present, third-generation Intel Core processors are not yet available in the inexpensive Celeron

variants. Consequently, a corresponding inexpensive implementation does not exist. And the AMD R-Series must first overcome design-in obstacles. So bearing this in mind, it will take a certain amount of time for inexpensive OEM implementations to appear on the market, unless alternative solutions are developed in the meantime with, for example, USB 3.0 bridges.

Current most cost-effective USB 3.0 implementations

This can be achieved, for example, with Celeron processors of the second-generation Intel Core (code-named Sandy Bridge) and an additional NEC bridge for USB 3.0. Using a method like this, solutions ready for serial production have been realised since the beginning of 2011. One such solution was a COM Express type 6-based computer-on-module (Fig. 1). It offers developers a fast time-to-market for this highly attractive interface. This design with USB 3.0 is one of the first embedded implementations which customers have been able to use before the introduction of chipsets that support USB3.0 and has thus enabled them to accelerate time-

to-market of USB 3.0 platforms.

The importance of such early availability of latest technologies is especially crucial for innovative companies, as when the chipsets are launched, end-user awareness and the demand for USB 3.0 increase suddenly. This gives a real competitive edge to OEMs who want to deliver USB 3.0 implementations parallel to chipset launches. That is possible only if you already have this type of technology ready for mass production.

First customer implementations with USB 3.0

A first customer using the module, which works with a USB 3.0 bridge on a solution with Intel Celeron processor of the 'Sandy Bridge' generation, will present a first, highly integrated solution at the Vision 2012 which uses cost-efficient USB 3.0 camera modules. Another customer is currently developing a traffic surveillance solution and a third is creating a box PC for SFF vision applications, to name just three examples in the vision segment.

All these applications are projects that promise technological leadership in the corresponding segment—for example, in combination with M2M solutions. Their competitive advantage time-wise—in comparison to chipset-based solutions—is at least the design-in phase or probably even longer, assuming that in future one wants to opt for using Ivy Bridge-based Intel Celeron processors.

USB 3.0 technology platforms for today and tomorrow

OEMs who did not start designing at the beginning of 2011, now, of course, have a much wider choice. To start using this technology cost-efficiently, today, they will nevertheless either have to use the Celeron version with USB 3.0 bridge, as it will be a cost-efficient